

$$1.a) \begin{cases} x + y - z = 2 \\ 2x + y - 3z = 2 \\ x - y - z = 0 \end{cases}$$

$$A^e = \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 2 & 1 & -3 & 2 \\ 1 & -1 & -1 & 0 \end{array} \right) \xrightarrow{\substack{L_2 = L_2 - 2L_1 \\ L_3 = L_3 - L_1}} \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & -1 & -1 & -2 \\ 0 & -2 & 0 & -2 \end{array} \right) \xrightarrow{L_2 = L_2 \cdot (-1)}$$

$$= \left(\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & 1 & 2 \\ 0 & -2 & 0 & -2 \end{array} \right) \xrightarrow{\substack{L_1 = L_1 - L_2 \\ L_3 = L_3 + 2L_2}} \left(\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 2 & 2 \end{array} \right) \xrightarrow{\substack{L_1 = L_1 + 2L_2 \\ L_3 = L_3 / 2}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right)$$

$$= \left(\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right) \xrightarrow{\substack{L_1 = L_1 - 2L_3 \\ L_2 = L_2 - L_3}} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right) \Rightarrow \begin{cases} x = 2 \\ y = 1 \\ z = 1 \end{cases}$$

$$b) \begin{cases} x + y + 2z = 1 \\ x + y - 3z = 1 \\ x + y - 2z = 1 \end{cases}$$

S_1 nu are det $\Rightarrow S'_1 = \{(2, 1, 1)\}$

$$A^e = \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 1 & 1 & 3 & 1 \\ 1 & 1 & -2 & 1 \end{array} \right) \xrightarrow{\substack{L_2 = L_2 - L_1 \\ L_3 = L_3 - L_1}} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -4 & 0 \end{array} \right)$$

avem 0 pe al doilea pivot $\Rightarrow$ sistemul este incompatibil (nu are solutie). $\Rightarrow S'_2 = \emptyset$

$$c) \begin{cases} x + y + z + t = 1 \\ 2x - y + z - t = 2 \\ x - 2y + 0z - 2t = -1 \end{cases}$$

$$A^e = \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 2 & -1 & 1 & -1 & 2 \\ 1 & -2 & 0 & -2 & -1 \end{array} \right) \xrightarrow{\substack{L_2 = L_2 - 2L_1 \\ L_3 = L_3 - L_1}} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & -3 & -1 & -3 & 0 \\ 0 & -3 & -1 & -3 & -2 \end{array} \right)$$

$$\xrightarrow{L_2 = \frac{L_2}{-3}} \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & \frac{1}{3} & 1 & 0 \\ 0 & -3 & -1 & -3 & -2 \end{array} \right) \xrightarrow{\substack{L_1 = L_1 - L_2 \\ L_3 = L_3 + 3L_2}} \left(\begin{array}{cccc|c} 1 & 0 & \frac{2}{3} & 0 & 1 \\ 0 & 1 & \frac{1}{3} & 1 & 0 \\ 0 & 0 & \underline{0} & \underline{0} & -2 \end{array} \right)$$

al treilea pivot 0 nu fie pe coloana termenilor liberi $\Rightarrow$
sistemul S_3 este incompatibil $S_3 = \emptyset$

$$d) \begin{cases} x + 2y + 3z - 2t = 6 \\ 2x - y - 2z - 3t = 8 \\ 3x + 2y - z + 2t = 4 \\ 2x - 3y - 2z + t = -8 \end{cases}$$

$$A^e = \left(\begin{array}{cccc|c} 1 & 2 & 3 & -2 & 6 \\ 2 & -1 & -2 & -3 & 8 \\ 3 & 2 & -1 & 2 & 4 \\ 2 & -3 & 2 & 1 & -8 \end{array} \right) \xrightarrow{\substack{L_2 = L_2 - 2L_1 \\ L_3 = L_3 - 3L_1 \\ L_4 = L_4 - 2L_1}} \left(\begin{array}{cccc|c} 1 & 2 & 3 & -2 & 6 \\ 0 & -5 & -8 & -7 & -4 \\ 0 & -4 & -10 & 8 & -14 \\ 0 & -7 & -4 & 5 & -20 \end{array} \right)$$

$$\xrightarrow{L_2 = \frac{L_2}{-5}} \left(\begin{array}{cccc|c} 1 & 2 & 3 & -2 & 6 \\ 0 & 1 & \frac{8}{5} & -\frac{7}{5} & -\frac{4}{5} \\ 0 & -4 & -10 & 8 & -14 \\ 0 & -7 & -4 & 5 & -20 \end{array} \right) \xrightarrow{\substack{L_1 = L_1 - 2L_2 \\ L_3 = L_3 + 4L_2 \\ L_4 = L_4 + 7L_2}} \left(\begin{array}{cccc|c} 1 & 0 & -\frac{1}{5} & -\frac{8}{5} & \frac{22}{5} \\ 0 & 1 & \frac{8}{5} & -\frac{7}{5} & -\frac{4}{5} \\ 0 & 0 & -\frac{2}{5} & \frac{18}{5} & -\frac{28}{5} \\ 0 & 0 & \frac{36}{5} & \frac{18}{5} & -\frac{72}{5} \end{array} \right)$$

$$\begin{aligned} L_3 &= L_3 - \frac{5}{9} \\ L_4 &= L_4 - \frac{5}{16} \end{aligned} \quad \begin{pmatrix} 1 & 0 & \frac{1}{5} & -\frac{8}{5} & \frac{22}{5} \\ 0 & 1 & \frac{8}{5} & \frac{1}{5} & \frac{4}{5} \\ 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 2 & 1 & -4 \end{pmatrix} \quad \begin{aligned} L_1 &= L_1 + \frac{L_3}{5} \\ L_2 &= L_2 - L_3 \cdot \frac{8}{5} \\ L_4 &= L_4 - 2L_3 \end{aligned}$$

$$\begin{pmatrix} 1 & 0 & 0 & -2 & 5 \\ 0 & 1 & 0 & 3 & -4 \\ 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 5 & 10 \end{pmatrix} \quad L_4 = \frac{L_4}{5} \quad \begin{pmatrix} 1 & 0 & 0 & -2 & 5 \\ 0 & 1 & 0 & 3 & -4 \\ 0 & 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 1 & 2 \end{pmatrix} \quad \begin{aligned} L_1 &= L_1 + 2L_4 \\ L_2 &= L_2 - 3L_4 \\ L_3 &= L_3 - L_4 \end{aligned}$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -2 \end{pmatrix} \Rightarrow \begin{cases} x = 1 \\ y = 2 \\ z = -1 \\ t = -2 \end{cases}$$

S_4 is complete $\Rightarrow S_4 = \{(1, 2, -1, -2)\}$

$$S_1 = \begin{cases} x + y - z = 0 \\ 2x + y - 3z = 0 \\ x - y - z = 0 \end{cases}$$

$$A^T = \begin{pmatrix} 1 & 1 & -1 & 0 \\ 2 & 1 & -3 & 0 \\ 1 & -1 & -1 & 0 \end{pmatrix} \quad \begin{aligned} L_2 &= L_2 - 2L_1 \\ L_3 &= L_3 - L_1 \end{aligned} \quad \begin{pmatrix} 1 & 1 & -1 & 0 \\ 0 & -1 & -1 & 0 \\ 0 & -2 & 0 & 0 \end{pmatrix} \quad L_2 = -L_2$$

$$\begin{pmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -2 & 0 & 0 \end{pmatrix} \quad \begin{aligned} L_1 &= L_1 - L_2 \\ L_3 &= L_3 + 2L_2 \end{aligned} \quad \begin{pmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 \end{pmatrix} \quad L_3 = \frac{L_3}{2}$$

$$= \begin{pmatrix} 1 & 0 & -2 & | & 6 \\ 0 & 1 & 1 & | & 0 \\ 0 & 0 & 1 & | & 0 \end{pmatrix} \xrightarrow{\substack{L_1 = L_1 + 2L_3 \\ L_2 = L_2 - L_3}} \begin{pmatrix} 1 & 0 & 0 & | & 0 \\ 0 & 1 & 0 & | & 0 \\ 0 & 0 & 1 & | & 0 \end{pmatrix}$$

S_1 is comp det $\Rightarrow S'_1 = \{(0, 0, 0)\}$

b) $S_2 \begin{cases} x + y - z + t = 0 \\ x - y + z + t = 0 \\ 2x - y - 2z - t = 0 \end{cases}$

$$A^e = \begin{pmatrix} 1 & 1 & -1 & 1 & | & 0 \\ 1 & -1 & 1 & 1 & | & 0 \\ 2 & 1 & 2 & -1 & | & 0 \end{pmatrix} \xrightarrow{\substack{L_2 = L_2 - L_1 \\ L_3 = L_3 - 2L_1}} \begin{pmatrix} 1 & 1 & -1 & 1 & | & 0 \\ 0 & -2 & 2 & 0 & | & 0 \\ 0 & -1 & 4 & -3 & | & 0 \end{pmatrix}$$

$$\xrightarrow{L_2 = \frac{L_2}{-2}} \begin{pmatrix} 1 & 1 & -1 & 1 & | & 0 \\ 0 & 1 & -1 & 0 & | & 0 \\ 0 & -1 & 4 & -3 & | & 0 \end{pmatrix} \xrightarrow{\substack{L_1 = L_1 - L_2 \\ L_3 = L_3 + L_2}} \begin{pmatrix} 1 & 0 & 0 & 1 & | & 0 \\ 0 & 1 & -1 & 0 & | & 0 \\ 0 & 0 & 3 & -3 & | & 0 \end{pmatrix}$$

$$\xrightarrow{L_3 = \frac{L_3}{3}} \begin{pmatrix} 1 & 0 & 0 & 1 & | & 0 \\ 0 & 1 & -1 & 0 & | & 0 \\ 0 & 0 & 1 & -1 & | & 0 \end{pmatrix} \xrightarrow{L_2 = L_2 + L_3} \begin{pmatrix} 1 & 0 & 0 & 1 & | & 0 \\ 0 & 1 & 0 & -1 & | & 0 \\ 0 & 0 & 1 & -1 & | & 0 \end{pmatrix}$$

x, y, z var principale

$t = \alpha, \alpha \in \mathbb{R}$ var sc

S_2 is comp red $\Rightarrow \begin{cases} x = -\alpha \\ y = \alpha \\ z = \alpha \end{cases}$

$\Rightarrow S'_2 = \{(-\alpha, \alpha, \alpha, \alpha) \mid \alpha \in \mathbb{R}\}$

$$b. a) A = \left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 \end{array} \right) \xrightarrow{L_1 \leftrightarrow L_2} \left(\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array} \right) \xrightarrow{L_2 = L_2 - 2L_1}$$

$$= \left(\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 0 & -5 & 1 & -2 \end{array} \right) \xrightarrow{L_2 = \frac{L_2}{-5}} \left(\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 0 & 1 & -\frac{1}{5} & \frac{2}{5} \end{array} \right) \xrightarrow{L_1 = L_1 - 3L_2}$$

$$= \left(\begin{array}{cc|cc} 1 & 0 & \frac{3}{5} & -\frac{1}{5} \\ 0 & 1 & -\frac{1}{5} & \frac{2}{5} \end{array} \right) \Rightarrow A^{-1} = \left(\begin{array}{cc} \frac{3}{5} & -\frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{array} \right)$$

$$b) B = \left(\begin{array}{ccc|ccc} 2 & -1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 1 & 0 \\ 1 & 1 & -1 & 0 & 0 & 1 \end{array} \right) \xrightarrow{L_1 \leftrightarrow L_2} \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 0 & 1 & 0 \\ 2 & -1 & 1 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 & 0 & 1 \end{array} \right) \xrightarrow{L_2 = L_2 - L_1, L_3 = L_3 - L_1}$$

$$= \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 0 & 1 & 0 \\ 0 & -3 & -3 & 1 & -2 & 0 \\ 0 & 0 & -3 & 0 & -1 & 1 \end{array} \right) \xrightarrow{L_2 = \frac{L_2}{-3}} \left(\begin{array}{ccc|ccc} 1 & 1 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & -3 & 0 & -1 & 1 \end{array} \right) \xrightarrow{L_1 = L_1 - L_2}$$

$$= \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{3} & -\frac{1}{3} & 0 \\ 0 & 1 & 1 & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & -3 & 0 & -1 & 1 \end{array} \right) \xrightarrow{L_3 = -L_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & \frac{1}{3} & -\frac{1}{3} & 0 \\ 0 & 1 & 1 & -\frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 3 & 0 & 1 & -1 \end{array} \right) \xrightarrow{L_1 = L_1 - L_3, L_2 = L_2 - L_3}$$

$$= \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{3} & -\frac{2}{3} & 1 \\ 0 & 1 & 0 & -\frac{1}{3} & -\frac{1}{3} & 1 \\ 0 & 0 & 1 & 0 & 1 & -1 \end{array} \right) \Rightarrow B^{-1} = \left(\begin{array}{ccc} \frac{1}{3} & -\frac{2}{3} & 1 \\ -\frac{1}{3} & -\frac{1}{3} & 1 \\ 0 & 1 & -1 \end{array} \right)$$

$$1) C = \left(\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 2 & 1 & -2 & 0 & 1 & 0 \\ 1 & -1 & 4 & 0 & 0 & 1 \end{array} \right) \begin{array}{l} \\ L_2 = L_2 - 2L_1 \\ L_3 = L_3 - L_1 \end{array} = \left(\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 0 & -1 & -8 & -2 & 1 & 0 \\ 0 & -2 & 1 & -1 & 0 & 1 \end{array} \right)$$

$$\begin{array}{l} \\ L_1 = -L_2 \end{array} = \left(\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & 8 & 2 & -1 & 0 \\ 0 & -2 & 1 & -1 & 0 & 1 \end{array} \right) \begin{array}{l} \\ \\ L_1 = L_1 - L_2 \\ L_3 = L_3 + 2L_2 \end{array} = \left(\begin{array}{ccc|ccc} 1 & 0 & -5 & -1 & 1 & 0 \\ 0 & 1 & 8 & 2 & -1 & 0 \\ 0 & 0 & 17 & 3 & -2 & 1 \end{array} \right)$$

$$\begin{array}{l} \\ L_3 = \frac{L_3}{17} \end{array} = \left(\begin{array}{ccc|ccc} 1 & 0 & -5 & -1 & 1 & 0 \\ 0 & 1 & 8 & 2 & -1 & 0 \\ 0 & 0 & 1 & \frac{3}{17} & -\frac{2}{17} & \frac{1}{17} \end{array} \right) \begin{array}{l} \\ \\ L_1 = L_1 + 5L_3 \\ L_2 = L_2 - 8L_3 \end{array}$$

$$= \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{2}{17} & \frac{7}{17} & \frac{5}{17} \\ 0 & 1 & 0 & \frac{10}{17} & -\frac{1}{17} & -\frac{8}{17} \\ 0 & 0 & 1 & \frac{3}{17} & -\frac{2}{17} & \frac{1}{17} \end{array} \right) \Rightarrow C^{-1} = \left(\begin{array}{ccc} -\frac{2}{17} & \frac{7}{17} & \frac{5}{17} \\ \frac{10}{17} & -\frac{1}{17} & -\frac{8}{17} \\ \frac{3}{17} & -\frac{2}{17} & \frac{1}{17} \end{array} \right)$$